

Ticket #002: Ping to server fails. Monitoring alerts show the server is not responding to ICMP echo requests.

Symptoms: Pings from the network fail.

Resolution:

1. Testing network connectivity from another machine on the local network to the affected server.

```
[alejandrorabadan@Mac ~ % ping -c 2 192.168.1.94
PING 192.168.1.94 (192.168.1.94): 56 data bytes
Request timeout for icmp_seq 0
^C
--- 192.168.1.94 ping statistics ---
2 packets transmitted, 0 packets received, 100.0% packet loss
```

Analysis: ICMP echo requests are transmitted, but no replies are received.

2. Testing SSH connectivity.

```
[alejandrorabadan@Mac ~ % ssh jay@192.168.1.94
[jay@192.168.1.94's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic aarch64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
```

Analysis: SSH connection is successful. This proves the server is reachable and online. The server is likely dropping ICMP requests due to a firewall rule or security configuration issue.

3. Inspecting the iptables on the server.

```
[jay@mainserver:~$ sudo iptables -L -n -v | grep icmp
    0      0 DROP      1  --  *      *      192.168.64.0/24      0.0.0.0/0      icmp type 8
    15  1260 DROP      1  --  *      *      192.168.1.0/24      0.0.0.0/0      icmp type 8
[jay@mainserver:~$ exit
```

Analysis: Firewall inspection shows an INPUT chain rule dropping ICMP echo requests (type 8) from the 192.168.1.0/24 network. Packet counters on this rule increment when ping tests are run from the local network, confirming that the firewall is blocking ICMP requests and triggering the monitoring alert.

4. After consulting with the simulated admin, the rule is supposed to allow ICMP echo requests from the local network while blocking them from all other sources. In production, the local network router's firewall typically drops echo requests from external IPs to internal servers. For the purpose of this lab and documenting troubleshooting methodology, the rule will be configured on the server itself, which also provides an additional layer of security.

5. Identifying the specific entry number for the faulty rule, deleting it and verifying its deletion.

```
[jay@mainserver:~$ sudo iptables -L INPUT --line-number -v -n
Chain INPUT (policy ACCEPT 55641 packets, 10M bytes)
num  pkts bytes target     prot opt in     out     source               destination
1    67412 12M cali-INPUT 0      -- *    *       0.0.0.0/0         0.0.0.0/0           /* cali:Cz_u1IQiXIMmKD
2     2388 297K KUBE-PROXY-FIREWALL 0      -- *    *       0.0.0.0/0         0.0.0.0/0           ctstate NEW /*
3     293K 61M KUBE-NODEPORTS 0      -- *    *       0.0.0.0/0         0.0.0.0/0           /* kubernetes heal
4     2388 297K KUBE-EXTERNAL-SERVICES 0      -- *    *       0.0.0.0/0         0.0.0.0/0           ctstate NEW
portals */
5     293K 61M KUBE-FIREWALL 0      -- *    *       0.0.0.0/0         0.0.0.0/0
6      15 1260 DROP        1      -- *    *       192.168.1.0/24    0.0.0.0/0           icmptype 8
[jay@mainserver:~$ sudo iptables -D INPUT 6
[jay@mainserver:~$ sudo iptables -L -n -v | grep icmp
[jay@mainserver:~$
```

6. Creating a firewall rule to allow ICMP echo requests from the local network (192.168.1.0/24), and a separate rule to drop echo requests originating from all other networks. Then verifying that both rules are correctly applied.

```
[jay@mainserver:~$ sudo iptables -I INPUT 6 -p icmp --icmp-type echo-request -s 192.168.1/24 -j ACCEPT
[jay@mainserver:~$ sudo iptables -A INPUT -p icmp --icmp-type echo-request -j DROP
[jay@mainserver:~$ sudo iptables -L INPUT --line-number -v -n
Chain INPUT (policy ACCEPT 57944 packets, 9438K bytes)
num  pkts bytes target     prot opt in     out     source               destination
1    320K 59M cali-INPUT 0      -- *    *       0.0.0.0/0         0.0.0.0/0           /* cali:Cz_u
2    4181 525K KUBE-PROXY-FIREWALL 0      -- *    *       0.0.0.0/0         0.0.0.0/0           cts
3    539K 106M KUBE-NODEPORTS 0      -- *    *       0.0.0.0/0         0.0.0.0/0           /* kuber
4    4181 525K KUBE-EXTERNAL-SERVICES 0      -- *    *       0.0.0.0/0         0.0.0.0/0
portals */
5    539K 106M KUBE-FIREWALL 0      -- *    *       0.0.0.0/0         0.0.0.0/0
6      0 0 ACCEPT      1      -- *    *       192.168.1.0/24    0.0.0.0/0           icmptype 8
7      0 0 DROP        1      -- *    *       0.0.0.0/0         0.0.0.0/0           icmptype 8
```

Analysis: Both entries are valid (entries 6 and 7)

7. Testing ping to server from another computer on the local network.

```
[alejandrorabadan@Mac ~ % ping -c 2 192.168.1.94
PING 192.168.1.94 (192.168.1.94): 56 data bytes
64 bytes from 192.168.1.94: icmp_seq=0 ttl=64 time=0.574 ms
64 bytes from 192.168.1.94: icmp_seq=1 ttl=64 time=0.461 ms

--- 192.168.1.94 ping statistics ---
2 packets transmitted, 2 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 0.461/0.517/0.574/0.057 ms
```

Analysis: Ping from the local network to the server is successful. Firewall properly configured.

Conclusion: Ping failures to the server were caused by a firewall rule blocking ICMP echo requests from the local network (192.168.1.0/24). SSH connectivity confirmed the server was online and reachable despite ICMP being blocked. Investigation of iptables revealed an offending INPUT chain rule dropping ICMP traffic. The rule was removed and replaced with updated firewall entries allowing ICMP from the local network while blocking other sources. After applying the changes, ping tests succeeded, confirming that the firewall was correctly reconfigured and normal network visibility was restored.